Directional Lifting & Modal Decomposition — Methodology

Authors: (Prepared for you)

Purpose: This document describes how to (A) lift visibility into direction-aware representations, (B) decompose measured light into physical modes (direct, diffuse, specular, multi-bounce, sunlight/sky, discrete lamps), and (C) extract robust features that feed topology and chirality inference pipelines.

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1. Executive summary

Real scenes rarely provide the ideal "fully-open" direct visibility cover. To robustly infer source topology and chirality from measured image data we therefore:

1. Lift visibility from sets $V(P) \subset L$ (source point sets for pixel P) into a directional space $L \times S^2$ (or discretized approximation). This preserves handedness information and thickens thin/projective visibility patches.■2. Decompose measured light into physical modes (direct point-like, area/diffuse, skylight, specular single-bounce, multi-bounce) using a combination of geometric tests, heuristic rules, and optionally learned classifiers.■3. Extract modality-specific features (per-pixel, per-source-sample, and per-direction) such as directness, DOA histograms, penumbra width, specularity, color temperature, temporal responsiveness, and polarization when available.■4. Build modality-specific lifted covers and compute weighted/thresholded co-visibility complexes. Use persistent homology with mode-weighting and epsilon-thickening to infer stable topological features and chirality cues.

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2. Definitions and notation

- Scene geometry: ambient Euclidean space with camera(s) and objects (occluders). The extended light source (or set of light sources) is denoted by $L \subset R^3$ (often parameterizable in 2D parameters such as an image patch, disk, ring, etc.).■- Pixel P: an imaging location (pixel center or small patch) that receives radiance from directions on the hemisphere.■- Visibility set $V(P) \subset L$: the subset of source points whose straight-line segment to P is unobstructed (direct-ray sense).■- Directional lifting: representation of visibility as subsets of the product $L \times S^2$: element (s,ω) means source point $s \in L$ contributes to P via incoming direction ω at P.■- Modes: classes of light-transport phenomena:■ - Direct point-like (DP): straight line from s to P with highly concentrated angular distribution (sun, filament, bulb).■ - Area/diffuse (AD): extended source with broad angular spread (lamps with diffusers, skylight). - Specular single-bounce (SS): illumination that reaches P after a single specular reflection. - Indirect / multi-bounce (MB): light dominated by multiple scattering.■ - Environmental (ENV): sky dome / global illumination with characteristic clear-sky or overcast distributions.

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3. Representation & discretization

3.1 Source sampling

- Sample m candidate points s_i on L (uniform or importance sampled by radiance priors). Typical m ranges: 256–1024 for moderate scenes.

3.2 Directional discretization

- Discretize the incoming direction space at each pixel by either:■ - A set of spherical bins (e.g., 32–128 equal-area bins), or■ - Low-order spherical harmonic (SH) coefficients (orders 0–4) or zonal basis, or■ - A small mixture of von-Mises-Fisher components (parametric DOA model).

- Represent lifted visibility as a sparse incidence list of tuples (s_i, b_k) where b_k is a direction bin.

3.3 Modal layers

- Maintain separate incidence matrices (pixel × (s_i, direction_bin)) per mode: direct, area, specular, multi-bounce. Each entry stores a small vector of statistics (counts, radiance estimate, mean direction confidence).

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4. Feature extraction (what to compute and why)

Features are grouped by type: per-pixel, per-source-sample, per-(source,direction) tuple, and temporal/spectral/polarimetric enhancements.

4.1 Per-pixel features

- Directness / visible-fraction D(P): Monte Carlo estimate of fraction of sampled source points with unobstructed straight-line visibility to pixel P. Useful to identify pixels dominated by direct rays.

- DOA histogram / angular concentration (kappa): angular distribution of incident radiance. High kappa indicates point-like or specular contributions; low kappa indicates diffuse/skylight.

- Penumbra width / edge profile: compute local edge profiles across shadow boundaries; fit logistic or sigmoid curves to estimate width and curvature. Narrow, sharp edges -> small angular source; wide smooth edges -> extended/diffuse source.

- Specularity score: local normalized high-pass energy, highlight color matching to dominant source color, and sharpness metric.

- Temporal responsiveness: from a small burst or controlled jiggle, compute cross-correlation between camera/light motion and intensity changes per pixel. Strong responsiveness points to direct/single-bounce components.

- Color temperature (CCT) / spectral centroid: quick estimate of dominant spectral character of incoming radiance indicating sun/sky vs tungsten vs LED.

- Polarization signature (if available): degree and angle of polarization — helps separate sky/diffuse vs specular components.

4.2 Per-source-sample features (for each sampled s_i)

- Visibility support count: number of pixels (or fraction) that see s_i — histogram over i reveals coverage holes.

- Radiance estimate & variance: aggregated radiance from those pixels when mapped to s_i (radiometric inversion required) — indicates which parts of L are bright vs dark.

- Direction-to-camera distribution: how angles from s_i to different camera pixels distribute; used to build lifted elements.

4.3 Per-(source,direction) features

- Directional confidence: number of independent pixels/frames supporting the tuple (s_i, b_k).

- Mode likelihood vector: soft labels (direct, diffuse, specular, multi-bounce) computed per tuple using heuristics or a learned model.

- Geometric consistency score: if depth is available, raycasts verify unobstructed geometry for straight-line modes; for specular modes, reflect ray geometry through candidate mirror surfaces to test feasibility.

4.4 Temporal/spectral/polarimetric augmentations

- Flicker signatures: 50/60 Hz or PWM signs indicate mains-powered indoor lighting.■- Motion transfer: how highlights move relative to camera or light shifts — distinguishes specular re-projection versus diffuse changes.

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5. Modal decomposition — pipeline

Two complementary approaches: a heuristic/physics rule-based pipeline (fast, interpretable) and a learned pipeline (more robust, data-driven).

5.1 Heuristic / physics-driven pipeline

1. Compute base features (Section 4) from input frames (single HDR frame + short burst recommended).■2. Detect candidate direct points: pixels with high DOA kappa, narrow penumbra width, high HDR highlight intensity and stable color temperature -> seed direct point-like mode.■3. Detect diffuse/area regions: broad DOA distribution, low specularity score, and smooth spatial neighborhoods -> seed area/diffuse mode.■4. Detect specular reflections: pixels with sharp highlights that move according to reflectance geometry and respond strongly to small motions -> seed single-bounce/specular mode.■5. Detect multi-bounce / indirect: low directness but spatially coherent low-frequency responses, high variance under viewpoint change -> seed multi-bounce.■6. Cluster directional histograms across pixels to separate multiple sources of same mode (k-means, spectral clustering on histogram similarity).■7. Construct lifted incidence: for each mode and cluster produce incidence lists (s_i, b_k) with weights (confidence, radiance estimate).■8. Postprocessing: prune low-confidence tuples; smooth by neighborhood in L and direction space; apply epsilon-thickening where thin sets occur.

5.2 Learning-based pipeline

- Train a model to predict per-pixel mode probabilities and DOA histograms. Input stacks: RGB (HDR), local gradients/penumbra maps, temporal stack (small motion), depth (optional), polarimetric channels (optional).■- Loss terms: classification loss for modes, KL divergence for DOA histograms, geometric consistency loss (when ray geometry available).■- Synthetic data generation (Blender/Cycles) is recommended to bootstrap training: render many scenes with known L, occluders, materials, and sensor noise models. Label ground-truth lifted tuples and modes.

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6. Using the lifted / modal cover for topology and chirality

1. Primary topology inference: use the direct mode lifted cover as the primary open-like cover; build co-visibility complexes on sampled source points s_i and compute persistent homology.■2. Mode-weighted complexes: form graphs/simplicial complexes where edges and higher simplices are weighted by mode confidence (direct high weight, multi-bounce low weight). Use weighted persistent homology or thresholded filtrations.■3. Directional lifting advantage: many thin/projective visibility sets become volumetric in $L \times S^2$; compute nerve/persistent homology in the lifted space or projective back to L after epsilon-thickening.■4. Chirality detection: use specular single-bounce and directional signatures — reflect the lifted family to produce an enantiomeric candidate and test for isomorphism of oriented complexes. Asymmetries after reflection signal chirality in embedding.

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7. Algorithms and pseudocode (high-level)

7.1 Heuristic modal decomposition + lift (outline)

Input: image(s) (HDR or burst), optional depth, sampling grid on L (s_i)■Output: incidence lists per-mode: I_mode = {(s_i,dir_bin): weight, stats}

1. Preprocess: HDR merge, denoise, compute local gradients, detect shadow edges■2. Sample candidate source points s_i on L■3. For each pixel P:■ - estimate DOA histogram (via sampling s_i->P geometry or local steerable filters)■ - compute features: penumbra_width, DOA_kappa, spec_score, temporal_resp, CCT■ - assign soft mode probabilities p_mode(P)■4. Cluster DOA histograms to identify discrete source clusters per mode■5. For each (s_i, dir_bin): aggregate evidence from pixels P supporting that tuple■ - compute weight = sum_P p_mode(P) * direction_kernel(P.dir, dir_bin) * radiance_transfer(P,s_i)■6. Prune low-weight tuples, smooth in (s,dir) space, epsilon-thicken if needed■7. Return I_mode

7.2 Co-visibility and topology (brief)

- Build pixel × (s_i) incidence boolean/sparse matrix for direct mode (or weighted by confidence).■- Build co-visibility graph on s_i: connect s_i and s_j if exist P with both visible (optionally weight by number of co-visible pixels).■- Compute clique complex or VR complex and persistent homology (ripser/GUDHI) to estimate Betti numbers.■- For lifted space, build complexes on tuples or project back with thickening.

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8. Implementation notes & data structures

- Sparse incidence representation: store tuples (mode, s_index, dir_bin, pixel_indices[], weight, radiance_estimate) in a compact table.■- BVH or GPU raycaster: for geometric verification of direct visibility using depth or synthetic geometry.■- Direction kernels: use von-Mises-Fisher kernel to soften bin assignment.■- Parallelization: per-pixel and per-(s_i) computations are embarrassingly parallel — use GPU or multi-threaded workers.■- Topology libs: ripser.py (fast), GUDHI, Dionysus. For weighted complexes, consider converting weights to filtrations.

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9. Sensor recommendations & capture protocols

- Minimum: HDR single-shot + short burst (3–8 frames) with small hand jiggle (for temporal responsiveness). Depth optionally from device or stereo.■- Recommended additions: polarizer filter or polarimetric sensor, short baseline camera array (3–5 cameras) for coverage, spectral bands for CCT estimation.■- Capture protocol: one HDR frame for radiometric baseline, then a 5–8 frame burst with small lightweight motion (hand jiggle) and optionally a controlled light wiggle if possible.

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10. Experiments & validation plan

1. Synthetic unit tests: render canonical shapes (disk, annulus, multi-lamp) with occluders; validate that direct-mode lifted complexes recover expected Betti numbers.■2. Indoor real tests: office with ceiling fixtures and window daylight; verify modal labels match ground truth and directness coverage suffices for topology.■3. Outdoor sun+sky tests: verify sun separation as DP and sky as AD; test chirality detection using specular reflections in a controlled mirrored object.■4. Ablation tests: remove polarimetry, HDR, or depth to quantify degradation.

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11. Appendix: parameter suggestions

- source samples m: 256–1024■- direction bins: 32–128■- penumbra width thresholds: tune per sensor; start with normalized width relative to patch size■- DOA angular concentration kappa thresholds: (high kappa > 10 indicates point-like; low kappa < 2 indicates diffuse) — tune empirically■- persistence filter: keep homology classes with lifetime > 3× median noise lifetimes (bootstrap noise)

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12. Next steps and notes

- If you want, I can:■ - add pseudocode converted into a runnable Python prototype (with synthetic scene generation), or■ - produce a Blender scene recipe to generate labeled training data, or■ - convert this document into a PDF for sharing (downloadable).

End of document.

Module: Motion–Visibility Toolkit — API & Methods (appendix)

This appendix organizes the directional-lifting + modal-decomposition pipeline into a clearly callable module with methods a worker can query or invoke. Methods are categorized by ease of computation (Cheap / Moderate / Heavy), described with inputs/outputs, complexity, and recommended usage patterns.

[The document continues with the Detailed method primer, KDS, Space–time, Morse/catastrophe, Optic flow, Persistent homology sections, edge cases, implementation ease summary, pseudocode and tooling recommendations.]